

Computation of Multipliers Using Broyden's Method for a Generic Numerical Integrator

1 Objective

In many numerical methods, the advance from z_n to z_{n+1} depends on a multiplier or correction vector

$$\mu \in \mathbb{R}^q$$

that must be chosen so that the result satisfies a coupling, projection, or constraint condition. The computation of μ leads to a nonlinear system of the form

$$R_n(\mu) = 0, \quad R_n : \mathbb{R}^q \longrightarrow \mathbb{R}^q. \quad (1)$$

Broyden's method makes it possible to solve (1) without recomputing the exact Jacobian of the residual at every iteration. Its formulation depends only on the ability to evaluate $R_n(\mu)$ and not on the internal structure of the integrator. Therefore, it can be used with ABBA, BM4, or any other numerical method.

2 Generic Construction of the Residual

We separate the procedure into four maps.

2.1 Input map or lifting

The physical state $z_n \in \mathbb{R}^d$ and the multiplier $\mu \in \mathbb{R}^q$ are transformed into the state on which the numerical method acts:

$$Y_n(\mu) = \mathcal{E}_n(z_n, \mu) \in \mathbb{R}^m. \quad (2)$$

The map $\mathcal{E}_n$ may duplicate the state, shift some of its components, or introduce auxiliary variables.

2.2 Base integrator

Let

$$\Psi_{h,n} : \mathbb{R}^m \longrightarrow \mathbb{R}^m \quad (3)$$

be the one-step map of the chosen integrator. The state obtained after applying the method is

$$Y_{n+1}(\mu) = \Psi_{h,n}(Y_n(\mu)) = \Psi_{h,n}(\mathcal{E}_n(z_n, \mu)). \quad (4)$$

The notation $\Psi_{h,n}$ represents, in an abstract way, any scheme: ABBA, BM4, another composition method, or even an integrator not based on compositions.

2.3 Compatibility condition

The result must satisfy a condition written through

$$\mathcal{C}_n : \mathbb{R}^m \times \mathbb{R}^q \longrightarrow \mathbb{R}^q. \quad (5)$$

The associated residual is defined by

$$R_n(\mu) := \mathcal{C}_n(\Psi_{h,n}(\mathcal{E}_n(z_n, \mu)), \mu). \quad (6)$$

Thus, finding the multiplier is equivalent to solving

$$R_n(\mu_n) = 0.$$

The function $\mathcal{C}_n$ may represent, among other possibilities, the equality between several copies of the same variable, a constraint condition $g(z) = 0$, or a projection equation.

2.4 Reconstruction of the physical state

Once μ_n has been found, the physical state is reconstructed through a map

$$z_{n+1} = \mathcal{P}_n(Y_{n+1}(\mu_n), \mu_n), \quad (7)$$

where $\mathcal{P}_n$ depends on the specific architecture of the method.

In this way, the part specific to each integrator is contained in

$$\mathcal{E}_n, \quad \Psi_{h,n}, \quad \mathcal{C}_n, \quad \mathcal{P}_n,$$

whereas Broyden's algorithm remains unchanged.

3 Exact Jacobian of the Residual

Newton's method applied to (1) would be

$$\mu^{(k+1)} = \mu^{(k)} - K(\mu^{(k)})^{-1} R_n(\mu^{(k)}), \quad (8)$$

where

$$K(\mu) := D_\mu R_n(\mu) \in \mathbb{R}^{q \times q}. \quad (9)$$

If all maps are differentiable, the chain rule applied to (6) gives

$$\begin{aligned} K(\mu) = & D_Y \mathcal{C}_n(Y_{n+1}(\mu), \mu) D\Psi_{h,n}(Y_n(\mu)) D_\mu \mathcal{E}_n(z_n, \mu) \\ & + D_\mu \mathcal{C}_n(Y_{n+1}(\mu), \mu). \end{aligned} \quad (10)$$

This is the Jacobian of the nonlinear problem used to compute μ . It should not be confused with the physical Jacobian of the step $z_n \mapsto z_{n+1}$.

Computing (10) exactly at every iteration may be expensive because it contains the derivative of the entire integrator, $D\Psi_{h,n}$. Broyden's method replaces this matrix by an approximation that is updated using successive evaluations of the residual.

4 Broyden's Method

Suppose that the current iterate $\mu^{(k)}$, the residual

$$R_k := R_n(\mu^{(k)}) \quad (11)$$

and an approximation

$$K_k \simeq D_\mu R_n(\mu^{(k)})$$

are known.

The correction $\Delta\mu_k$ is obtained by solving

$$K_k \Delta\mu_k = -R_k, \quad (12)$$

and the multiplier is updated as

$$\mu^{(k+1)} = \mu^{(k)} + \Delta\mu_k. \quad (13)$$

The complete integrator is then evaluated again in order to compute

$$R_{k+1} := R_n(\mu^{(k+1)}).$$

Define

$$s_k := \mu^{(k+1)} - \mu^{(k)}, \quad y_k := R_{k+1} - R_k. \quad (14)$$

The vector s_k is the change introduced in the multiplier, and y_k is the observed change in the residual. The new Jacobian approximation must incorporate this information through the secant condition

$$K_{k+1} s_k = y_k. \quad (15)$$

The good Broyden update is given by

$$K_{k+1} = K_k + \frac{(y_k - K_k s_k) s_k^\top}{s_k^\top s_k}. \quad (16)$$

Indeed,

$$\begin{aligned} K_{k+1} s_k &= K_k s_k + \frac{(y_k - K_k s_k) s_k^\top s_k}{s_k^\top s_k} \\ &= y_k. \end{aligned}$$

Therefore, (16) is a multidimensional generalization of the secant method. Moreover, it is the smallest modification of K_k in the Frobenius norm that satisfies (15).

5 Initialization

The choice of the initial values depends on the specific problem, not on Broyden's algorithm itself.

5.1 Initial value of the multiplier

A common choice is

$$\mu^{(0)} = 0, \quad (17)$$

when μ represents a small correction and the uncorrected method provides a sufficiently close approximation. This choice does not state that $\mu = 0$ is the solution; it only fixes the first point at which R_n is evaluated.

The multiplier from the previous time step may also be used,

$$\mu^{(0)} = \mu_{n-1},$$

or any available predictor. In every case, Broyden's method starts by evaluating the corresponding residual exactly.

5.2 Initial matrix

The matrix K_0 should approximate

$$D_\mu R_n(\mu^{(0)}).$$

It can be obtained in different ways:

1. by evaluating the exact Jacobian only at the beginning;
2. by using an asymptotic approximation valid for small h ;
3. by finite differences, computing its columns as

$$K_0 e_j \approx \frac{R_n(\mu^{(0)} + \varepsilon e_j) - R_n(\mu^{(0)})}{\varepsilon}; \quad (18)$$

4. by using a simple constant matrix, for example $K_0 = \alpha I_q$, when the structure of the problem justifies this approximation.

The choice $K_0 = 4I_q$ is not part of Broyden's method in general. It is an initialization specific to certain formulations with two symmetric copies. For a different lifting, a different compatibility condition, or even a different residual, the corresponding approximation must be used.

6 Formulation Using the Approximate Inverse

Instead of storing K_k , one may directly store

$$H_k \simeq K_k^{-1}.$$

The iteration can then be written as

$$\mu^{(k+1)} = \mu^{(k)} - H_k R_k. \quad (19)$$

The inverse update associated with (16) is

$$H_{k+1} = H_k + \frac{(s_k - H_k y_k) s_k^\top H_k}{s_k^\top H_k y_k}, \quad (20)$$

provided that the denominator is nonzero. The initial matrix must satisfy

$$H_0 \simeq K_0^{-1}.$$

7 Generic Algorithm for One Time Step

To advance from z_n to z_{n+1} , the following procedure is applied:

1. Choose $\mu^{(0)}$ and an initial approximation K_0 .
2. For $k = 0, 1, 2, \dots$, construct

$$Y_n^{(k)} = \mathcal{E}_n(z_n, \mu^{(k)}).$$

3. Perform one complete step of the chosen integrator:

$$Y_{n+1}^{(k)} = \Psi_{h,n}(Y_n^{(k)}).$$

4. Evaluate the residual

$$R_k = \mathcal{C}_n(Y_{n+1}^{(k)}, \mu^{(k)}).$$

5. If

$$\|R_k\|_2 \leq \text{tol},$$

stop the iterations and set $\mu_n = \mu^{(k)}$.

6. Solve

$$K_k \Delta \mu_k = -R_k$$

and update

$$\mu^{(k+1)} = \mu^{(k)} + \Delta \mu_k.$$

7. Run the integrator again with $\mu^{(k+1)}$, evaluate R_{k+1} , and compute

$$s_k = \mu^{(k+1)} - \mu^{(k)}, \quad y_k = R_{k+1} - R_k.$$

8. Update

$$K_{k+1} = K_k + \frac{(y_k - K_k s_k) s_k^\top}{s_k^\top s_k}$$

and continue the iteration.

After convergence, the physical state is reconstructed as

$$z_{n+1} = \mathcal{P}_n(Y_{n+1}(\mu_n), \mu_n).$$

During all internal iterations, the state z_n remains fixed. Each new value of $\mu^{(k)}$ requires rerunning the integrator because both $Y_n(\mu)$ and $Y_{n+1}(\mu)$ depend on the multiplier. It is useful to distinguish the indices

$n = \text{time step}, \quad k = \text{internal Broyden iteration.}$
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8 Specialization to Different Integrators

8.1 ABBA scheme with two copies

In the usual ABBA formulation,

$$\mathcal{E}_n(z_n, \mu) = \begin{pmatrix} z_n + \mu \\ z_n - \mu \end{pmatrix}, \quad \Psi_{h,n} = \Phi_{\text{ABBA}},$$

and, if

$$\begin{pmatrix} u^{(2)}(\mu) \\ v^{(2)}(\mu) \end{pmatrix} = \Phi_{\text{ABBA}} \begin{pmatrix} z_n + \mu \\ z_n - \mu \end{pmatrix},$$

the compatibility condition is

$$\mathcal{C}_n \left(\begin{pmatrix} u^{(2)} \\ v^{(2)} \end{pmatrix}, \mu \right) = u^{(2)} - v^{(2)} + 2\mu.$$

Therefore,

$$R_n(\mu) = u^{(2)}(\mu) - v^{(2)}(\mu) + 2\mu.$$

When $h = 0$, Φ_{ABBA} is the identity and

$$R_n(\mu) = 4\mu, \quad K_0 = 4I_q$$

is a natural initial approximation for small h . After convergence,

$$z_{n+1} = u^{(2)}(\mu_n) + \mu_n = v^{(2)}(\mu_n) - \mu_n.$$

8.2 BM4 scheme

To use Broyden's method with BM4, it is enough to replace the base integrator by

$$\Psi_{h,n} = \Phi_{\text{BM4}}.$$

If BM4 uses the same two-copy lifting and the same compatibility condition, the maps $\mathcal{E}_n$, $\mathcal{C}_n$, and $\mathcal{P}_n$ remain unchanged. If the BM4 architecture uses different auxiliary variables or a different final condition, only these maps need to be redefined. The iterations (12)–(16) remain exactly the same.

8.3 Any other method

For a new integrator, it is only necessary to specify:

$$\boxed{\begin{aligned} Y_n(\mu) &= \mathcal{E}_n(z_n, \mu), \\ Y_{n+1}(\mu) &= \Psi_{h,n}(Y_n(\mu)), \\ R_n(\mu) &= \mathcal{C}_n(Y_{n+1}(\mu), \mu), \\ z_{n+1} &= \mathcal{P}_n(Y_{n+1}(\mu_n), \mu_n). \end{aligned}} \tag{21}$$

Once the function R_n has been defined, Broyden's method acts as a nonlinear solver independent of the integrator used to evaluate it.

9 Conclusion

The fundamental object for Broyden's method is not the integrator, but the residual $R_n(\mu)$. ABBA, BM4, or any other method enters only through the evaluation of this residual. Consequently, a modular implementation should separate

$$\boxed{\text{evaluation of } R_n(\mu) \quad \text{and} \quad \text{solution of } R_n(\mu) = 0 \text{ using Broyden's method.}}$$

This separation makes it possible to reuse the same Broyden solver with different integrators without modifying its update formulas.